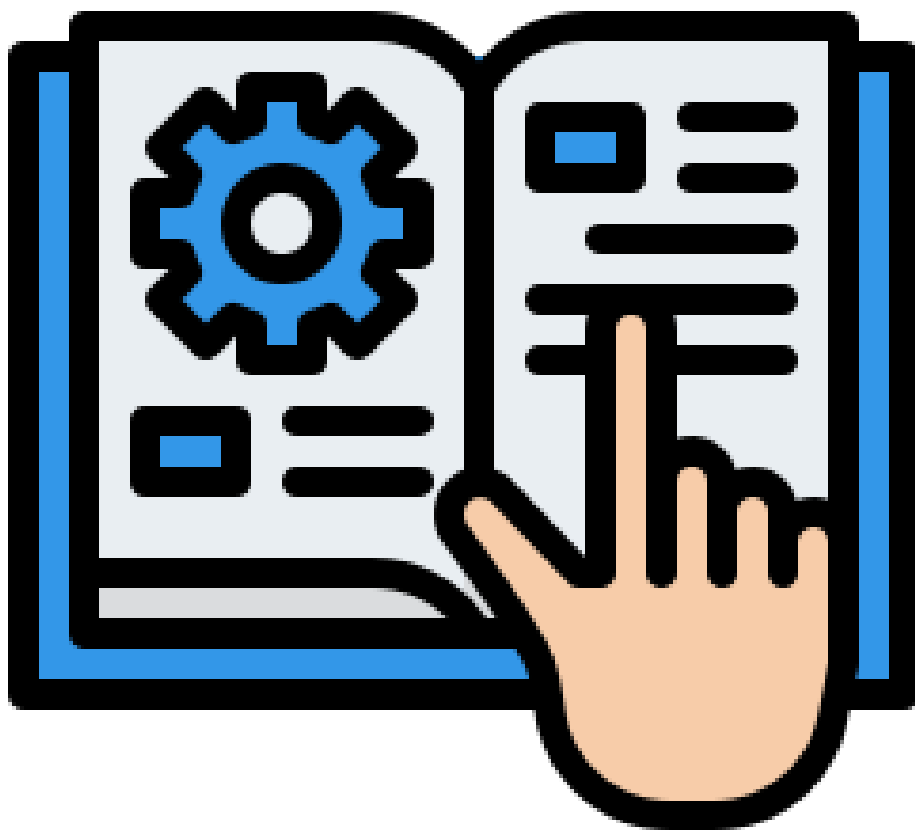


TECHNICAL MANUAL



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BUT Informatique 1
05/2026

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Introduction

This manual describes the steps required to install and configure a Debian 13 server in a virtual machine. It provides all commands needed for preparing the virtualization environment, installing the Debian operating system, and setting up Apache HTTP Server, PHP, and PostgreSQL services. The document also explains the configuration of network access, SSH access, and the administration tools required for proper server operation.

Debian 13 Server Installation

1. Installation Preparation (ISO Image)

First, we will download an ISO image. To do this, enter the URL into a search engine :

<https://cdimage.debian.org/cdimage/release/current/amd64/iso-cd/>

Then we select the file we need. For our installation, we will use the SHA512SUMS file. By clicking on the file at this URL :

<https://cdimage.debian.org/cdimage/release/current/amd64/iso-cd/SHA512SUMS>

We see 3 ISO image lines. We will use the first ISO image, whose name is:

debian-13.4.0-amd64-netinst.iso

Place it in a folder created for this purpose.

To do this, in your terminal, go to your directory using the following command :

- **cd DIRECTORY_NAME/PATH**

Download the Debian 13.4 ISO image :

- **wget <https://cdimage.debian.org/debian-cd/current/amd64/iso-cd/debian-13.4.0-amd64-netinst.iso>**

Verify the download :

- **ls -lh debian-13.4.0-amd64-netinst.iso**

To check that the ISO image is intact, in the terminal type the command :

sha512sum FILE_NAME

Example :

```
goyetcam@iut2-dg035-d03:~$ sha512sum /usr/local/images-ISO/debian-13.4.0-amd64-netinst.iso
```

Result of the command :

```
3e02de4ed744799350bd4039b137053835238ff9f9f29eee812309cd7eebdb5127bof2b54d167b76d3245  
30aa16939b41ae2d2f2d1995a2801e19938acdc927f /usr/local/images-ISO/debian-13.4.0-amd64-netinst.iso
```

Then you can visually compare the fingerprint you wrote in the document with the one from this command. (They are identical)

2. Installing the Debian system

Now that the ISO image is ready, we are going to prepare the machine using the installation ISO image. Use the command provided for this project : **S2.03-lance-installation**

Be careful : once you have run the command, you must immediately go to the next page to correctly follow the instructions. You can come back to the table later.

Example :

```
goyetcam@iut2-dg035-d03:~$ S2.03-lance-installation
```

The script is designed to prepare and start a Debian virtual machine using QEMU and KVM. The goal is to understand how the virtual machine is created and started. Its main role is to manage the virtual machine ISO image, avoid conflicts, and prepare the QEMU launch command.

Here are the parameters found in the QEMU command line process :

Paramètre	Rôle
qemu-system-x86_64	Run a 64-bit x86 VM
-machine q35	Modern virtual machine (Q35 chipset)
-cpu host	Use the host's real CPU
-m 4G	4 GB of RAM
-enable-kvm	Enable KVM hardware acceleration
-device VGA,xres=1024,yres=768	VGA graphics at 1024×768
-display gtk,zoom-to-fit=off	GTK graphical window
-drive ...	Configure the virtual disk
format=raw	Disk in raw format
file=\$image	Disk image file used
discard=unmap	Enable TRIM/UNMAP
-device e1000,netdev=net0	Intel E1000 network card
-netdev user,id=net0	NAT user networking
hostfwd=tcp::2222-:22	SSH forwarding
hostfwd=tcp::4443-:443	HTTPS forwarding (port 4443)
hostfwd=tcp::8080-:80	HTTP forwarding (port 8080)
hostfwd=tcp::5432-:5432	PostgreSQL forwarding (port 5432)

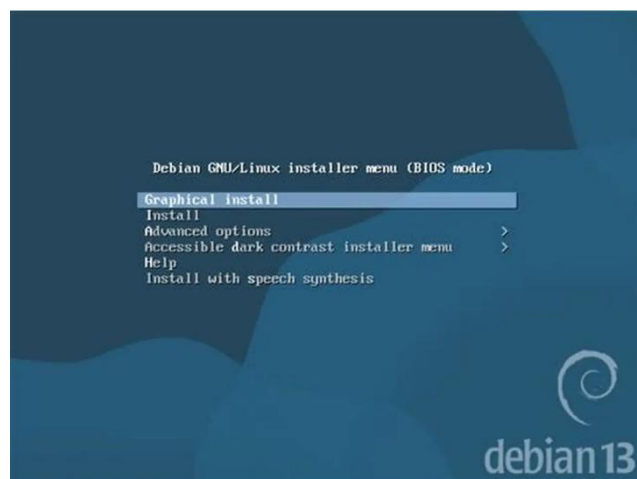
Debian installation setup :

When the first window appears, select Install
(See image)

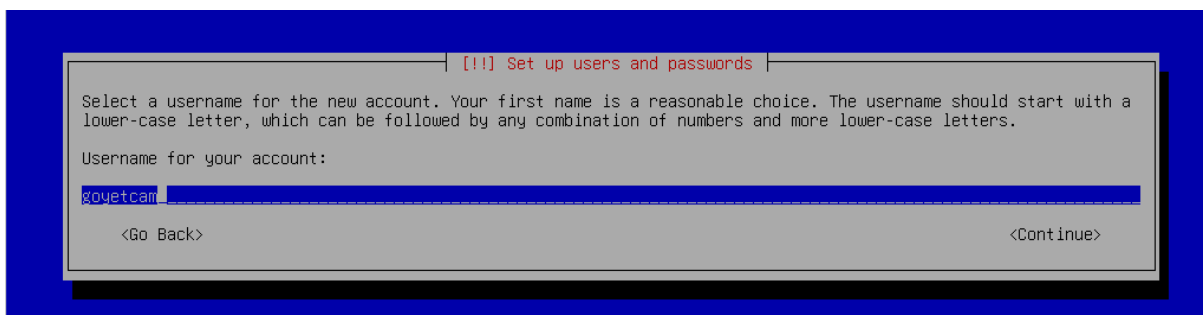
Then follow the steps one by one:

If nothing is specified, choose the default option.

- Language : English
- **Location : other/Europe/France**
- Locales : United States, en_US.UTF-8
- **Keyboard : French**
- **Hostname** : use server-"YOUR_UGA_LOGIN"
- **Root Password** : Enter a simple password, for example "root". Check the "Show Password" box to make sure the password you typed is the one you want.
- User Account - **Full Name** : your full name, for example "Camille Goyet"
- **User Name** : enter your UGA login username

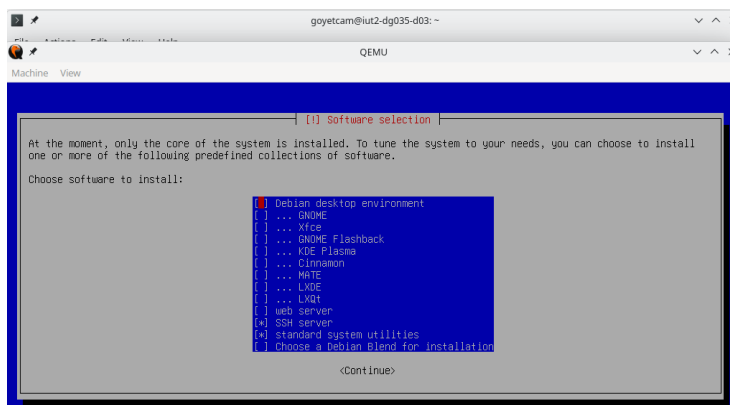


Screenshot of the virtual machine's first interface



Screenshot of the Debian installer during login entry

- **User Password** : enter a simple password, for example "etu". Check the "Show Password" box again to make sure the password is correct.
- Partition disks : Guided - use entire disk
- Partition disks : All files in one partition
- **Partition disks : Yes**
- Software Selection : Make sure that "Debian desktop" is not selected and that "SSH server" is selected.



Screenshot of the Debian installer at the "Software Selection" step

- Install GRUB : Yes
- **Device for boot loader** : /dev/sda

Once the installation is finished, the virtual machine restarts. You must shut it down before moving to the next step. Log in to the root account and shut it down, to do this :

With the commands inside the virtual machine :

- **root** -> To log in as root (an administrator account)
- **ROOT_PASSWORD** -> enter your root password

If the hash sign (#) is present, it means you are logged in as the root user.

Run this command to shut down the virtual machine:

- **# systemctl poweroff**
-> You must use this command every time you want to close the virtual machine.

This is the command you should use to properly shut down your virtual machine after each use.

Moving the disk image

The file was created on the local disk of the Linux workstation. You must move this image to the Erebus server so that you can use your virtual machine more easily later. You can also copy this image to a USB drive.

When the virtual machine is turned off, run this command in your terminal :

- **S2.03-déplace-image-disque-sur-erebus**

Example :

```
goyetcam@iut2-dg035-d03:~$ S2.03-déplace-image-disque-sur-erebus
```

3. Debian server check

To continue with the Debian server verification, we are going to start the virtual machine. In your terminal, type the command : **S2.03-lance-machine-virtuelle**

Write down the Ethernet and IP information of your machine and check that you can reach the outside network using the command in your machine's terminal(not the virtual machine):

With the commands :

- **\$ ipconfig**
- **\$ ping -c 9 DESTINATION**

Or, for an IUT computer user :

- **\$ ping -c 9 iut2-dgxx-dxx**

Example :

- goyetcam@iut2-dg035-d03:~\$ ipconfig
- goyetcam@iut2-dg035-d03:~\$ ping -c 9 iut2-dg035-d03

Result of the command :

- Ethernet characteristics: MAC address :
->52:54:00:12:34:56
- IP characteristics: IPv4 address:
->10.0.2.15/24
- IPv6 address : fec0::177c:71ec:4af8:3f

Log in as root (command seen previously).

Check that there is no graphical interface installed on your machine by running the command inside your virtual machine :

For X11, use this command :

- **# dpkg -l | grep libx11**

For Wayland, use this command :

- **# dpkg -l | grep wayland**

If nothing appears for both, skip the purge step. However, if files appear, run the following command : **apt purge NOM_DU_PACKAGE**

Example :

- **# apt purge libx11-6:amd64**
- **# apt purge libx11-data**

Now the machine does not have a graphical interface because this is done either to save resources, improve stability, increase security, or allow remote administration (SSH).

4. Port forwarding and SSH access

To access the simple user account of your virtual machine via SSH from your terminal.

Command from the Linux workstation :

- **\$ ssh VIRTUAL_MACHINE_NAME@localhost -p 2222**
- **su**
-> allows you to switch to the root user
- **apt install micro**
-> To test if everything is working

Example :

- **goyetcam@iut2-dgo35-d03:~\$ ssh goyetcam@localhost -p 2222**
- **su**
- **apt install micro**

5. Shutting down and restarting the virtual machine

If you made mistakes during the installation of your virtual machine, it is possible to delete the virtual machine and start again from scratch. To do this, you need to :

- Find the path of your virtual machine
- Go to the folder
- Delete the file using the command: **\$ rm FILE_NAME**
- Check that the deletion was successful using the command: **\$ ls -al**
- Run the installation command: **\$ S2.03- lance-installation**
- Perform the installation again by following the document from the beginning

Example :

- goyetcam@iut2-dg035-d03:~\$ ls -al /users/Stockage-HDD/images-kvm/S2.03/images/Debian-S2.03-goyetcam.img
- goyetcam@iut2-dg035-d03:~\$ rm /users/Stockage-HDD/images-kvm/S2.03/images/Debian-S2.03-goyetcam.img
- goyetcam@iut2-dg035-d03:~\$ S2.03-lance-installation

If you want to stop here for now, don't forget to properly shut down your machine using the command :

- **# systemctl poweroff**
->(inside your virtual machine, as the root user).

Installing Apache, PostgreSQL and PHP

If you have closed your virtual machine, start it again to perform the following installations (command in your terminal if you forgot : **S2.o3-lance-machine-virtuelle**).

The command :

- **# cat FILE_NAME**

This command is used to display the contents of a file.

```
root@serveur-goyetcam:~# cat /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options>        <dump> <pass>
# / was on /dev/sda1 during installation
UUID=0f0f58d4-2349-4afa-aa05-8ca816e637d3 /          ext4      errors=remount-ro 0    1
# swap was on /dev/sda5 during installation
UUID=9013dc01-72d5-440d-93bf-633d84b28c3a none      swap      SW        0    0
/dev/sr0    /media/cdrom0   udf,iso9660 user,noauto 0      0
root@serveur-goyetcam:~#
```

Screenshot of the output of the command cat /etc/fstab in your virtual machine

1.Install Apache

Now, we are going to install and check whether Apache is present on our virtual machine. To do this, follow the commands below inside your virtual machine as the root user.

The command :

- **# apt install apache2**
->install apache 2
- **# systemctl start apache2**
->start apache2
- **# systemctl status apache2**
-> check that apache2 is running

The screenshot shows two terminal windows. The left window displays the output of the command `systemctl status apache2`, showing that the service is loaded and active (running). The right window displays the output of the command `systemctl status ssh`, showing that the service is also loaded and active (running).

Screenshot of the output of the systemctl status command for SSH, Apache, and

Each screenshot shows the different installations. PostgreSQL installation will be done later.

```
root@serveur-goyetcam:~# systemctl status postgresql
● postgresql.service - PostgreSQL RBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2026-05-12 10:36:05 CEST; 12min ago
  Invocation: 611476f4f8884776b8e44241b11bdca5
     Process: 793 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 793 (code=exited, status=0/SUCCESS)
       Mem peak: 1.8M
          CPU: 4ms

May 12 10:36:05 serveur-goyetcam systemd[1]: Starting postgresql.service - PostgreSQL RBMS...
May 12 10:36:05 serveur-goyetcam systemd[1]: Finished postgresql.service - PostgreSQL RBMS.
root@serveur-goyetcam:~#
```

We are going to test the Apache software. It is not possible to display an HTML page because our machine is a server without a graphical interface.

From your virtual machine, in root mode, run:

- **#netcat -zv localhost 80**

From your Linux workstation :

- **\$ netcat -4zv localhost 8080**

Example from your virtual machine :

- root@serveur-goyetcam:# netcat -zv localhost 80

Result :

- localhost [127.0.0.1] 80 (http) open

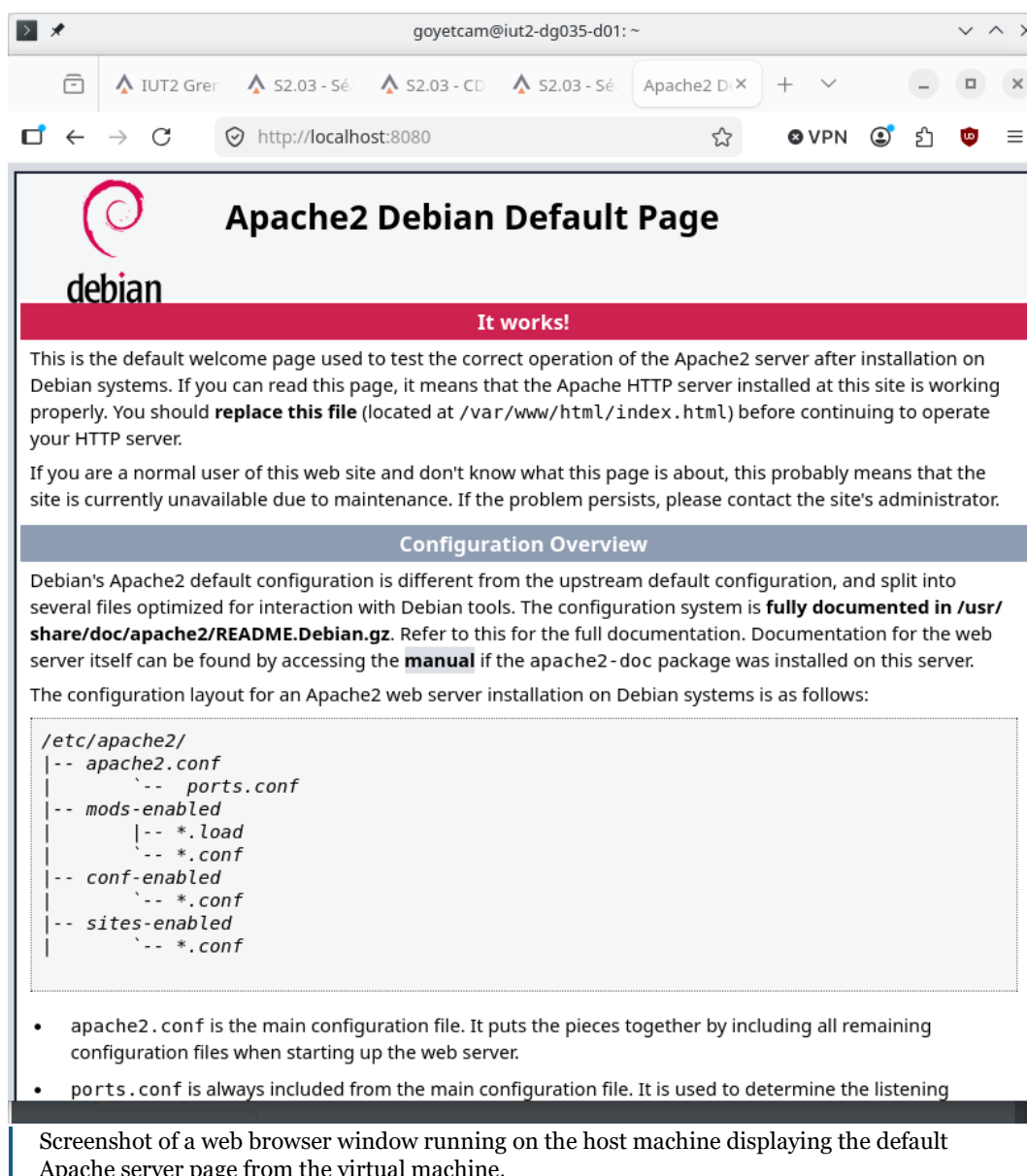
Example from your Linux workstation :

- goyetcam@iut2-dg035-d01:~\$ netcat -4zv localhost 8080

Result :

- Connection to localhost (127.0.0.1) 8080 port [tcp/http-alt] succeeded!

This is done by the provided launch script. You can therefore open a web browser on the host Linux machine and access the URL : `http://localhost:8080`.



3.Install PostgreSQL

Now, we move on to installing PostgreSQL.

Install the server and the client on your virtual machine using the following command as root:

Commands :

- **# apt install postgresql**
->install postgresQL
- **# su – postgres**
-> log in using the postgres usernam
- **\$ psql -l**
->display the list of databases

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
postgres	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres +
template0	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			postgres=CtC/postgres
template1	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres +
								postgres=CtC/postgres

(4 rows)

Screenshot showing the default list of databases.

We are now going to use PostgreSQL from the virtual machine:

1. Connect to PostgreSQL with: **# psql**
2. Create a user with a password :
create user NAME createrole createdb password 'PASSWORD'
3. Create a database whose owner is your user :
create database NAME_DATABASE owner USER_OWNER
-> The OWNER clause is used to specify the database owner. This user has full rights on the database and can create, modify, and delete tables.

Example from your virtual machine :

- postgres=# psql
- postgres=# create user goyetcam with createrole createdb password 'goyetcam';
- postgres=# create database basegoyetcam owner goyetcam;

```
postgres@serveur-goyetcam:~$ psql
psql (17.9 (Debian 17.9-0+deb13u1))
Type "help" for help.

postgres=# create user goyetcam with createrole createdb password 'goyetcam';
CREATE ROLE
postgres=# _
```

Screenshot showing an example of user creation

4. Connect to your created database from your PostgreSQL in your VM :

- **# psql**
 - **\$ psql NAME_DATABASE**
- postgres@serveur:\$ psql basegoyetcam
 - basegoyetcam=#

5. Create a table :

create table NAME_TABLE (COLUMN1 TYPE,..., COLUMNn TYPE) ;

6. Insert rows into your table :

Insert into NAME_TABLE values (data1,...data) ;

Example :

- basegoyetcam=# create table personne (id Numeric, nom VARCHAR, prenom VARCHAR);
- basegoyetcam=# insert into personne values (1,'goyet','camille'),(2,'blachon','lysiane');

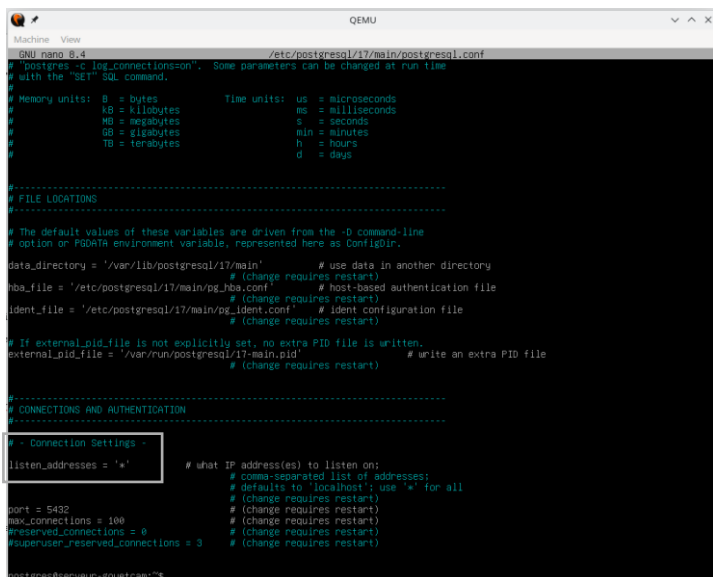
```
basegoyetcam=> create table personne (id NUMERIC, nom VARCHAR, prenom VARCHAR);
CREATE TABLE
basegoyetcam=> insert into personne values (1,'goyet','camille'),(2,'blachon','lysiane');
INSERT 0 2
basegoyetcam=> select * from personne;
 id |  nom  | prenom
-----+-----+-----
  1 | goyet  | camille
  2 | blachon | lysiane
(2 rows)
```

Screenshot of a PostgreSQL query executed from the command line on the virtual machine.

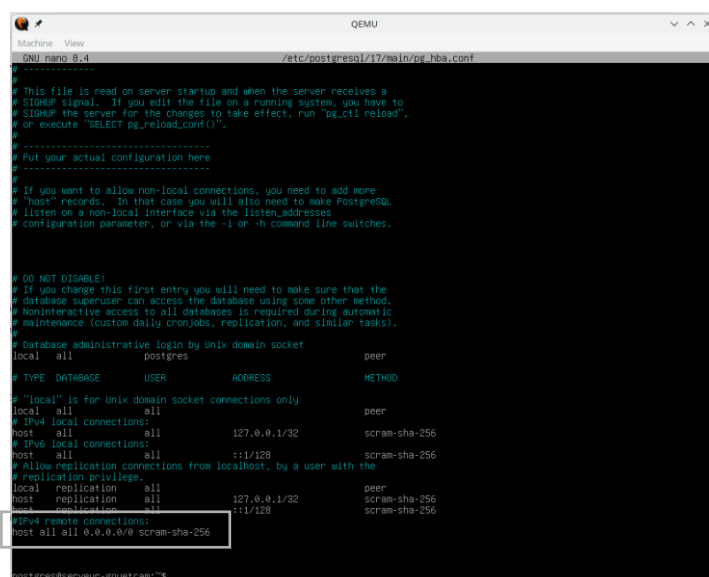
Now we are going to configure PostgreSQL so it can be accessed from your Linux workstation. To do this, you will need to modify two configuration files, then restart the PostgreSQL server.

Example :

- postgres@serveur:\$ **nano** /etc/postgresql/17/main/postgresql.conf
Fill in the file with:
 - **CONNECTION AND AUTHENTICATION** **listen_addresses='*'**
- postgres@serveur:\$ **nano** /etc/postgresql/17/main/pg_hba.conf
Fill in the file with:
 - **#IPv4 remote connections :**
 - **Host all all 0.0.0.0/0 scram-sha-256**
-> Ensure that only connections authenticated with a hashed password are allowed



Screenshot of the modification of the postgresql.conf file



Screenshot of the modification of the pg_hba.conf file

After completing these two configuration files, you need to exit PostgreSQL by typing: **exit**.

Run : root@serveur-goyetcam:# **systemctl restart postgresql.service**

In the Linux terminal, check that PostgreSQL is working correctly using a Linux command of your choice:

Example :

- goyetcam@iut2-dg035-d03:~\$ **Select * from personne ;**

```

goyetcam@iut2-dg035-d03: ~
File Actions Edit View Help
goyetcam@i...035-d03: ~ x goyetcam@i...035-d03: ~ x goyetcam@i...035-d03: ~ x goyetcam@i...035-d03: ~ x
goyetcam@iut2-dg035-d03:~$ psql -h localhost basegoyetcam
Password for user goyetcam:
psql (17.9 (Debian 17.9-0+deb13u1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, AL
PN: postgresql)
Type "help" for help.

goyetcam@basegoyetcam=> select * from personne;
 id |  nom  | prenom
-----+-----+-----
  1 | goyet | camille
  2 | blachon | lysiane
(2 rows)

goyetcam@basegoyetcam=> 

```

Screenshot of a PostgreSQL query executed from the command line on the Linux

```

commar goyetcam@iut2-dg035-d03: ~
Machine View
QEMU
postgres=# \l

```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
basegoyetcam	goyetcam	UTF8	libc	en_US.UTF-8	en_US.UTF-8			
postgres	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			
template0	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres +
template1	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres +
								postgres=Ctc/postgres

```

(4 rows)

```

Screenshot showing the list of PostgreSQL databases with their respective owners.

List the contents of the system table `pg_shadow` and verify that the passwords are correctly hashed using the SHA-256 hashing function.

```

QEMU
Machine View
root@serveur-goyetcam:~# systemctl restart postgresql.service
root@serveur-goyetcam:~# su - postgres
postgres@serveur-goyetcam:~$ psql
psql (17.9 (Debian 17.9-0+deb13u1))
Type "help" for help.

postgres=# select * from pg_shadow
postgres=# ;

```

username	usesysid	usecreatedb	usesuper	usebypassrls	passwd	valuntil	useconfig
goyetcam	16388	t	f	f	SCRAM-SHA-256\$4096:mEr1eF5eRcJ10AbWJcNU0g==:5aTVcEFVdt4L		

```

(2 rows)
(END)

```

Screenshot of a PostgreSQL query showing the system table `pg_shadow`, which should contain passwords hashed using the SHA-256 hashing function.

4.Install PHP

Now we move on to installing PHP.

Install it on your virtual machine using the following command as root:

Commands :

- **# apt install php-common libapache2-mod-php php-cli**
->install PHP
- **# /etc/init.d/apache2 stop**
- **# /etc/init.d/apache2 start**
-> Stopping and starting Apache once PHP is installed
- **# cd /var/www/html**
-> go to the html folder
- **# nano info.php**
-> create or open the file

In the empty window that opens, write:

- ```
<?php
phpinfo();
phpinfo(INFO_MODULES);
?>
```

| PHP Version 8.4.16                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| System                                                                                | Linux serveur-goyetcam 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Build Date                                                                            | Dec 18 2025 21:19:25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Build System                                                                          | Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Build Provider                                                                        | Debian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Server API                                                                            | Apache 2 Handler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Virtual Directory Support                                                             | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Configuration File (php.ini) Path                                                     | /etc/php/8.4/apache2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Loaded Configuration File                                                             | /etc/php/8.4/apache2/php.ini                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Scan this dir for additional .ini files                                               | /etc/php/8.4/apache2/conf.d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Additional .ini files parsed                                                          | /etc/php/8.4/apache2/conf.d/10-opcache.ini, /etc/php/8.4/apache2/conf.d/10-pdo.ini, /etc/php/8.4/apache2/conf.d/20-calendar.ini, /etc/php/8.4/apache2/conf.d/20-ctype.ini, /etc/php/8.4/apache2/conf.d/20-exif.ini, /etc/php/8.4/apache2/conf.d/20-ffi.ini, /etc/php/8.4/apache2/conf.d/20-fileinfo.ini, /etc/php/8.4/apache2/conf.d/20-ftp.ini, /etc/php/8.4/apache2/conf.d/20-gettext.ini, /etc/php/8.4/apache2/conf.d/20-iconv.ini, /etc/php/8.4/apache2/conf.d/20-phar.ini, /etc/php/8.4/apache2/conf.d/20-posix.ini, /etc/php/8.4/apache2/conf.d/20-readline.ini, /etc/php/8.4/apache2/conf.d/20-shmop.ini, /etc/php/8.4/apache2/conf.d/20-sockets.ini, /etc/php/8.4/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.4/apache2/conf.d/20-sysvsem.ini, /etc/php/8.4/apache2/conf.d/20-tokenizer.ini |
| PHP API                                                                               | 20240924                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| PHP Extension                                                                         | 20240924                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Zend Extension                                                                        | 420240924                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Zend Extension Build                                                                  | API420240924,NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PHP Extension Build                                                                   | API20240924,NTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| PHP Integer Size                                                                      | 64 bits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Debug Build                                                                           | no                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Thread Safety                                                                         | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Zend Signal Handling                                                                  | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Zend Memory Manager                                                                   | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Zend Multibyte Support                                                                | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Zend Max Execution Timers                                                             | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| IPv6 Support                                                                          | enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| DTrace Support                                                                        | disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Registered PHP Streams                                                                | https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Registered Stream Socket Transports                                                   | tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Registered Stream Filters                                                             | zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, convert.iconv.*                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Screenshot: a page showing the main characteristics of your PHP installation should appear :  
- <http://localhost:8080/info.php>

## 5.Install PhpPgAdmin

Now we move on to installing PHP.

Install it on your virtual machine using the following command as root:

Command :

- **# apt-get install phppgadmin**  
->install PHPPgAdmin

We are going to use commands to access a phpPgAdmin web page and run an SQL query using phpPgAdmin.

Commands :

- **# nano /etc/apache2/conf-available/phppgadmin.conf**  
-> create or open the file

In the empty window that opens, write:

- Look for the following setting:**Require local**
- Change it to:**Require all granted**

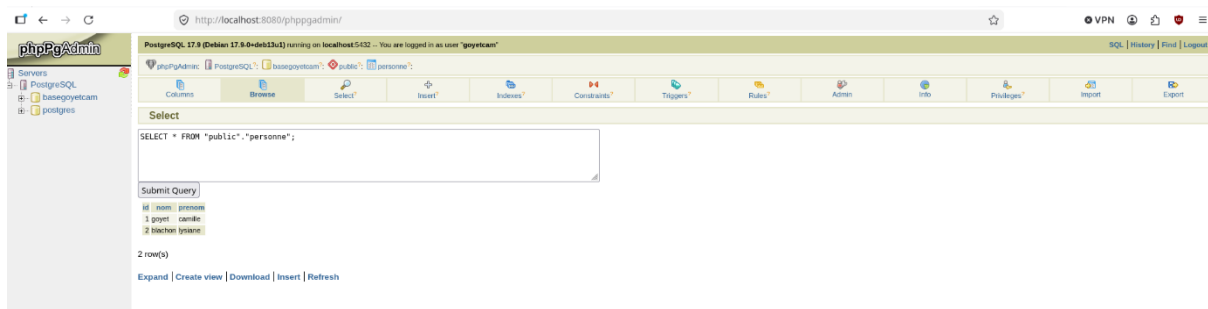
Save, exit, and switch to root mode in your virtual machine :

- After changing the setting, make sure that the syntax is correct : **apache2ctl -t**
- Make sure that the config file is enabled : **a2enconf phppgadmin**
- After changing the configuration restart the Apache service : **systemctl restart apache2**

On your machine, in a web browser, enter the following link :

- **http://localhost:8080/phppgadmin**

On the web page, look at the top left to go to **PostgreSQL**. Enter the database (**DATABASE\_NAME**) using the username created at the beginning. Continue entering the database, then the table. Then click on “**Browse**”. Finally, write your SQL query and click “**Submit Query**”.



Screenshot of a PostgreSQL query using phpPgAdmin



To be able to navigate to the webpage `page_sae_S2.03.php`, run :

### Commands :

- **# blkid**

-> execute on the virtual machine in root mode

Copy it to your Apache server on your virtual machine. To do this, in your terminal :

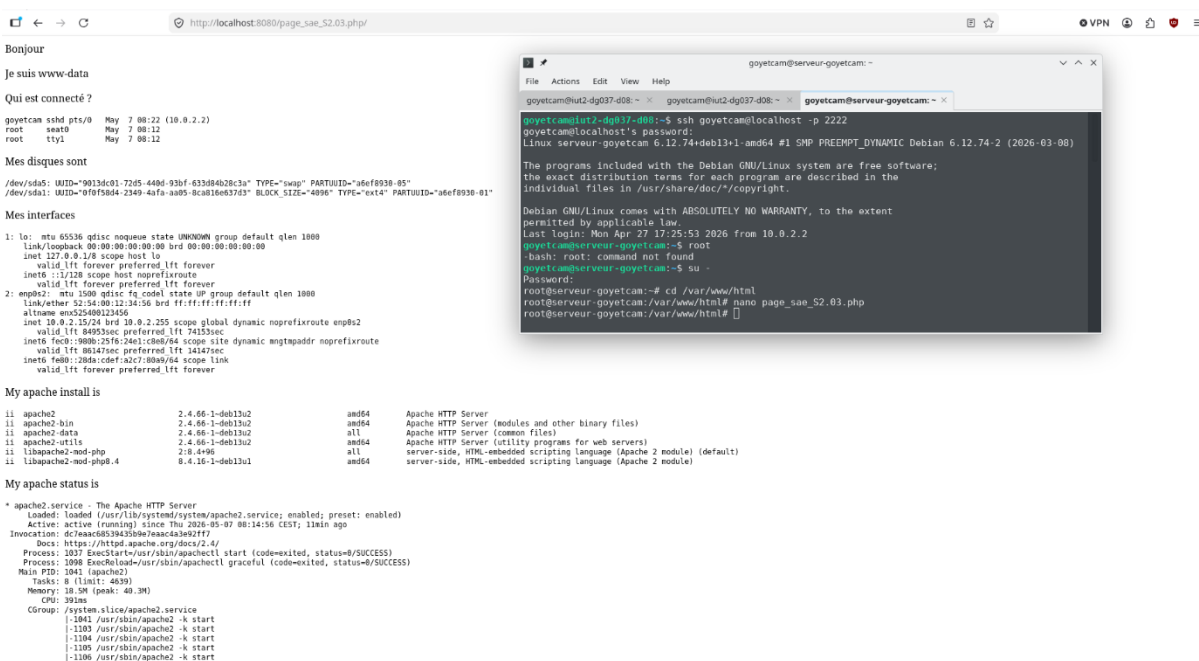
- Copy the content of this file:  
/users/info/www/intranet/enseignements/S2.03/page\_sae\_S2.03.php
- **\$ ssh MACHINE\_NAME -p 2222**
- **\$ su -**
- **# cd /var/www/html**
- **# nano page\_sae\_S2.03.php**
- Paste the text you copied previously

Go to your machine and open a web browser :

- [http://localhost:8080/page\\_sae\\_S2.03.php/](http://localhost:8080/page_sae_S2.03.php/)

### Example :

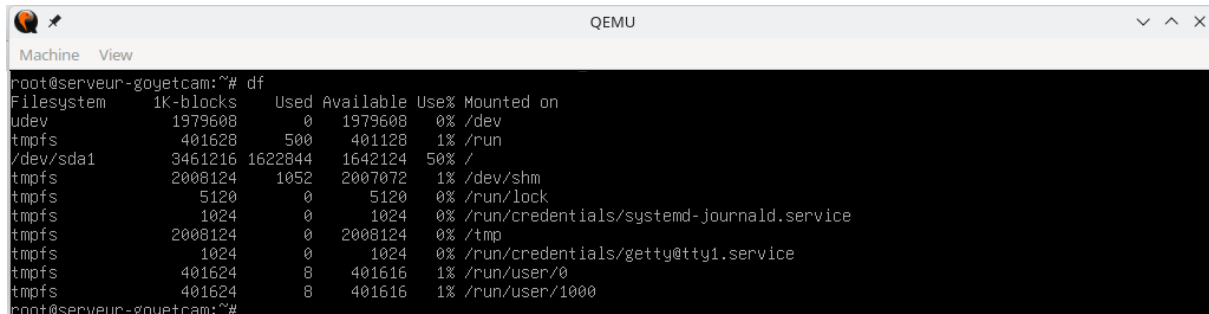
- `root@serveur-goyetcam: # blkid`
- `goyetcam@iut2-dg037-d08:~$ ssh goyetcam@localhost -p 2222`
- `goyetcam@serveur-goyetcam:~$ su -`
- `root@serveur-goyetcam:~# cd /var/www/html`
- `root@serveur-goyetcam:/var/www/html# nano page_sae_S2.03.php`



Screenshot of the host web browser window querying the PHP file



To show the storage space used and remaining on your virtual machine, run the following command inside the VM as root : **# df**

A screenshot of a QEMU terminal window. The window title is 'QEMU'. The terminal shows the command 'df' being executed. The output is a table with columns: Filesystem, 1K-blocks, Used, Available, Use%, and Mounted on. The data rows are: udev (1979608, 0, 1979608, 0%, /dev), tmpfs (401628, 500, 401128, 1%, /run), /dev/sda1 (3461216, 1622844, 1642124, 50%, /), tmpfs (2008124, 1052, 2007072, 1%, /dev/shm), tmpfs (5120, 0, 5120, 0%, /run/lock), tmpfs (1024, 0, 1024, 0%, /run/credentials/systemd-journald.service), tmpfs (2008124, 0, 2008124, 0%, /tmp), tmpfs (1024, 0, 1024, 0%, /run/credentials/getty@tty1.service), tmpfs (401624, 8, 401616, 1%, /run/user/0), and tmpfs (401624, 8, 401616, 1%, /run/user/1000). The prompt is 'root@serveur-goyetcam:~#'.

| Filesystem | 1K-blocks | Used    | Available | Use% | Mounted on                                |
|------------|-----------|---------|-----------|------|-------------------------------------------|
| udev       | 1979608   | 0       | 1979608   | 0%   | /dev                                      |
| tmpfs      | 401628    | 500     | 401128    | 1%   | /run                                      |
| /dev/sda1  | 3461216   | 1622844 | 1642124   | 50%  | /                                         |
| tmpfs      | 2008124   | 1052    | 2007072   | 1%   | /dev/shm                                  |
| tmpfs      | 5120      | 0       | 5120      | 0%   | /run/lock                                 |
| tmpfs      | 1024      | 0       | 1024      | 0%   | /run/credentials/systemd-journald.service |
| tmpfs      | 2008124   | 0       | 2008124   | 0%   | /tmp                                      |
| tmpfs      | 1024      | 0       | 1024      | 0%   | /run/credentials/getty@tty1.service       |
| tmpfs      | 401624    | 8       | 401616    | 1%   | /run/user/0                               |
| tmpfs      | 401624    | 8       | 401616    | 1%   | /run/user/1000                            |

Screenshot showing the used and remaining storage space on your virtual machine

## 6. A safety analysis of your installation

In all installations, there may be software security vulnerabilities that can appear. These may occur in Debian, Apache, PHP, or PostgreSQL.

To fix these vulnerabilities, we could have implemented a stronger password, updated packages, and installed security updates with specific configurations for the different applications.

Therefore, vulnerabilities are mainly fixed by keeping Debian and the software up to date using apt and security updates.



For more information, an explanatory video in French is available. The content of this section was taken from the video:

‘Configurer Connexion SSH sans Mot de Passe | Clé SSH-KEYGEN’ by Numelion IT.

## 2. Direct access to the root account via SSH

This final step consists of connecting via SSH to directly access root mode.

Command inside the virtual machine :

- **vi /etc/ssh/sshd\_config**  
-> edit the file : 'sshd\_config'  
Add a line in the '**Authentication**' section of the file indicating:  
**PermitRootLogin yes**. This line may already exist but be commented out with a #. In that case, remove the #.  
→ See the green box in the screenshot.
- Save the changes
- Restart the sshd service : **service sshd restart**

Command in your machine's terminal :

- **ssh root@localhost -p 2222**  
-> allows you to connect directly as root

```

goyetcam@iu...g037-d08: ~
File Actions Edit View Help
goyetcam@iu...g037-d08: ~ x goyetcam@iu...g037-d08: ~ x goyetcam@iu...g037-d08: ~ x goyetcam@iu...g037-d08: ~ x
goyetcam@iu...g037-d08: ~ x
The strategy used for options in the default sshd_config shipped with
OpenSSH is to specify options with their default value where
possible, but leave them commented. Uncommented options override the
default value.
Include /etc/ssh/sshd_config.d/*.conf
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
Ciphers and keying
#RekeyLimit default none
Logging
#SyslogFacility AUTH
#LogLevel INFO
Authentication:
#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
#PubkeyAuthentication yes
Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
#AuthorizedPrincipalsFile none
#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
root@serveur-goyetcam:~# sudo nano /etc/ssh/sshd_config
-bash: sudo: command not found
root@serveur-goyetcam:~# systemctl restart sshd

```

Screenshot showing the commands used to connect via SSH directly as root.

The steps in this section were found using this website; if you want more technical information :

<https://clients.offshore-cloud.com/knowledgebase/50/Activer-or-Desactiver-la-connexion-root-en-SSH.html>

## Commands Summary

| Command                       | Explanation                                            |
|-------------------------------|--------------------------------------------------------|
| apt install XXX               | Install a package                                      |
| apt purge PACKAGE_NAME        | Remove a package and its configuration files           |
| blkid                         | Display partition UUIDs and filesystem types           |
| cat FILE_NAME                 | Display file contents                                  |
| cd DIRECTORY_NAME/PATH        | Change current directory                               |
| chmod                         | Modify file or directory permissions                   |
| dpkg -l   grep libx11         | Check whether X11 libraries are installed              |
| dpkg -l   grep wayland        | Check whether Wayland packages are installed           |
| echo                          | Print text or write output to a file                   |
| exit                          | Exit the current shell or session                      |
| ip a / ip addr                | Display network interfaces and IP addresses            |
| ipconfig                      | Display network configuration on Windows               |
| ls -option                    | List directory contents with options                   |
| mkdir                         | Create a directory                                     |
| nano                          | Terminal-based text editor                             |
| netcat -zv localhost 80       | Check whether port 80 is open                          |
| ping -c 9 DESTINATION         | Send 9 ICMP echo requests to test network connectivity |
| psql -l                       | List PostgreSQL databases                              |
| rm FILE_NAME                  | Remove a file                                          |
| root                          | Superuser account with full system privileges          |
| sha512sum FILE_NAME           | Compute SHA-512 checksum of a file                     |
| ssh VM_NAME@localhost -p 2222 | Connect to the virtual machine using SSH on port 2222  |
| ssh-keygen                    | Generate an SSH key pair                               |
| su                            | Switch user account                                    |
| su - postgres                 | Switch to the PostgreSQL user account                  |
| systemctl poweroff            | Shut down the system                                   |
| systemctl restart XXX         | Restart a service                                      |
| systemctl start XXX           | Start a service                                        |
| systemctl status XXX          | Display service status information                     |
| vi                            | Terminal-based text editor                             |
| wget                          | Download files from the Internet                       |

## Conclusion

In this technical manual, you have seen the installation of Debian, as well as PostgreSQL, PHP, and phpPgAdmin. You have learned how to configure them in order to access their web pages from your Linux machine. Finally, you can now perform SSH access commands.